

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-FA-8901-01 Sheet 1 of 1
OPL Title	Level Gauge / Bridle Verification (LT-8901)	Discipline	Instrument	
Equipment	FA-8901 - REFLUX ACCUMULATOR DRUM	Area / Unit	8900 - HEXANE COLUMN OVERHEAD SYSTEM	
Related Interlock	SEQ-8901 (SIL 1)	P&ID; Ref	TJC-LLD-PID-8901	
Classification:	☒ Basic Knowledge	☐ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **level gauge / bridle verification (lt-8901)** for FA-8901 (REFLUX ACCUMULATOR DRUM) in the HEXANE COLUMN OVERHEAD SYSTEM. A false level reading causes reflux pump cavitation (low) or column flooding (high); the transmitter must agree with the gauge. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Hexane / water** at up to 10 barg / FV / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Cross-check LT-8901 against the level bridle gauge	Loop isolated / under permit
2	Blow down the bridle to clear any plugging	Reference standard traceable
3	Confirm the LSHH-8901 and LSLL-8901 switch actions	As-found & as-left recorded
4	Verify the level tracks correctly through a level change	Trip / control action verified
5	Log any deviation and recalibrate if needed	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Reflux drum level transmitter drifting vs gauge	Transmitter zero drift	Recalibrated LT-8901 against the bridle gauge
Blanket / vent split-range valve hunting	Split-range calibration overlap incorrect	Re-tuned PCV-8905 split-range, verified PT-8902 hold
Proof test of drum high-high level trip	Within tolerance	Recalibrated LSHH-8901 and verified EFF-1 column feed trip

6. KEY LEARNING POINTS

- A false level reading causes reflux pump cavitation (low) or column flooding (high); the transmitter must agree with the gauge.
- Respect the interlock SEQ-8901: e.g. LSHH-8901 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-8901 before working on FA-8901.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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